Building an Open-Source Web GIS Solution

Shahid Nawaz Khan
Assistant Professor
Institute of Geographical Information Systems
National University of Sciences and Technology (NUST)



NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

What is a Web GIS Solution?

- A system for creating, managing, analyzing, and visualizing geographical data over the web.
- Components typically include:
  1. A web server
  2. A spatial database
  3. A server-side application (backend)
  4. A client-side application (frontend)
- We will focus on building a completely open-source solution using popular and robust tools.



NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

The Open-Source Technology Stack

- The complete stack includes the frontend (mapping library), backend (web framework), and the spatial database.



NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

Frontend: The Mapping Library Leaflet

- A modern open-source JavaScript library for mobile-friendly interactive maps.
- Known for being lightweight and simple.
- Handles all the rendering on the client-side.
- Key functionalities:
  - Displaying base maps from various providers (e.g., OpenStreetMap).
  - Adding spatial data (markers, lines, polygons).
  - Managing map layers and controls.
  - Handling user interactions (clicks, zooms, drags).



NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

Backend: Python Framework (Django and GeoDjango)

- **Django:** A high-level Python web framework that encourages rapid development and clean, pragmatic design.
- **GeoDjango:** A bundled extension of Django that turns it into a world-class GIS platform.
- **Key Features for GIS:**
  - Spatial-enabled data models that map to database spatial types.
  - Powerful spatial lookup and query methods (e.g., ‘distance‘, ‘intersects‘).
  - Support for various spatial databases.
  - Integration with GIS libraries like GDAL/GEOS.



NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

Spatial Database: The Heart of the Data (PostGIS)

- An open-source extension for the PostgreSQL database.
- Provides spatial objects (geometries) to the database.
- Adds spatial functions and operators to manipulate and analyze spatial data.
- **Key Features:**
  - Standard spatial data types: ‘Point‘, ‘LineString‘, ‘Polygon‘, ‘MultiPolygon‘.
  - Spatial functions for analysis: ‘ST_Buffer‘, ‘ST_Intersects‘, ‘ST_Distance‘.
  - Spatial indexing using R-trees for fast query performance.

NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

User Authentication

- A critical component for any real-world application.
- Allows users to register, log in, and manage their data securely.
- **Django's Built-in Solution:**
  - Django provides a robust, pre-built authentication system out of the box.
  - It handles user sessions, password hashing, and permissions.
  - This saves a significant amount of development time and effort.



NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

The Complete Stack

- **Frontend:** HTML, CSS, and JavaScript with the **Leaflet** library.
- **Backend: Python** using the **Django** framework with the **GeoDjango** extension.
- **Database: PostgreSQL** with the **PostGIS** extension.



NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

Summary and Key Takeaways

- The combination of Django/GeoDjango (backend), PostGIS (database), and Leaflet (frontend) provides a powerful, flexible, and completely open-source stack.
- This stack offers a scalable solution for a wide range of Web GIS applications, from simple map viewers to complex spatial analysis tools.
- The open-source nature means no licensing costs and a large, active community for support.



NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

Questions?

Thank You!



NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY